Diabetes Prediction Using Machine Learning

Group 4 — ITAI 1371: Introduction to Machine Learning
Midterm Project Proposal

Project Overview

Our group selected the Diabetes Prediction Dataset from Kaggle for our machine learning project. The dataset contains 100,000 records with information such as age, gender, hypertension, heart disease, smoking history, BMI, HbA1c level, blood glucose level, and diabetes status.

Our goal is to prepare this dataset for developing machine learning models that can predict whether a patient has diabetes based on these health-related features.

Dataset Preparation

To make the dataset easier to process in Google Colab, we used stratified random sampling to reduce it from 100,000 to 50,000 records while maintaining the original class distribution of 91.5% without diabetes and 8.5% with diabetes.

We then divided the data into 35,000 training records (70%) and 15,000 testing records (30%). All exploratory data analysis and preprocessing were performed on the training data only.

Our preprocessing included investigating missing values and duplicate records, examining data distributions, encoding categorical features, standardizing numerical features, and creating a new feature called high_bmi.

We also used SMOTENC to address class imbalance. The resulting processed training dataset contains 64,050 records, with 32,025 records in each diabetes class.

Machine Learning Objective

For our final project, we plan to train and compare classification models, such as Logistic Regression, Decision Tree, and Random Forest, to predict diabetes status.

We will evaluate their performance using accuracy, precision, recall, F1-score, and confusion matrices. We will pay particular attention to recall for diabetes-positive cases because failing to identify a positive case can be important in a health-related prediction task.

Expected Outcome

We expect to produce a prepared dataset suitable for machine learning experiments and identify which classification model performs best on previously unseen testing data.

This project is intended for educational prediction analysis and not for medical diagnosis.

Dataset Source:

https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset